EDUHUB / ENGINEERING DOCUMENTATION

UML CLASS DIAGRAMS

EduHub | Online Learning Platform

38	15	03
ENTITY CLASSES	UML DIAGRAMS	N-TIER LAYERS

Scope: UML class structure, attributes and design-level operations, association multiplicities, inheritance, service contracts and design patterns.

Companion to the approved candidate ERD, physical schema, use cases, sequence diagrams and state machines.

Document status: proposed implementation-level class model, not a reverse-engineered C# codebase. Persistent entities match the prior 38-entity baseline.

SYSTEM ANALYSIS & DESIGN / NOVEMBER 2026

DOCUMENT ROADMAP

Object model and dependencies

Class diagrams should describe object structure and responsibilities, without replacing the ERD or a detailed API specification.

01 / Domain structure

11 domain views covering all 38 database entities, plus an overview.

02 / Application contracts

Controllers, service interfaces, external integrations and data-access abstractions.

03 / Design patterns

Repository + Unit of Work, Strategy and Observer with explicit realizations.

04 / Validation

Multiplicity, identity ownership, transaction responsibilities and cross-document checks.

38 entity classes / 15 class diagrams / 30 editable diagram files / no generated implementation code

NOTATION

How to read these diagrams

UML 2.5.1-style class boxes use three compartments: name, selected attributes, and representative operations.

ELEMENT	NOTATION / INTERPRETATION	EDUHUB EXAMPLE
Visibility	+ public; C# property types shown as string?, long, decimal.	+ CourseId: long {FK}
Class members	The displayed fields are selected; use Logical & Physical Schema for exhaustive columns.	+ MarkCompleted(): void
Association	Solid line and multiplicities specify links between instances.	Course 1 - 0..* Enrollment
Composition	Filled-diamond association indicates conceptual lifetime ownership, not SQL cascade delete.	Course 1 *-- 0..* CourseSection
Realization	Dashed line with hollow triangle: a class implements an interface.	StripeGateway implements IPaymentGateway
Generalization	Solid line with hollow triangle indicates inheritance / specialized interface.	ICourseRepository : IRepository<T>
Dependency	Dashed arrow means one class depends on an abstraction.	CheckoutController -> ICheckoutService
Design status	Entity attributes map to existing schema; methods/services are proposed contracts.	No existing code implied

BOUNDARIES

Exactly what is modeled

Preserve the relational model while adding object-oriented behavior and layer interfaces.

CLASS FAMILY	COUNT	OWNERSHIP / IMPLEMENTATION
ASP.NET Core Identity conceptual types	3	ApplicationUser, ApplicationRole, UserRole are mapped by Identity; four auxiliary framework tables remain outside the ERD count.
EduHub custom persistent entities	35	Entity Framework Core domain types representing tables from the companion SQL schema.
Controller / service / adapter samples	27	Proposed technical classes/interfaces; intentionally do not add tables.
Nested C# enums / value objects	Not counted	Represent SQL-backed status values and in-memory calculation parameters; no extra entities.
Total persistent model	38	8 domains and identical entity names to the ERD baseline.

Design decision: Keep the required EduHub.Web, EduHub.Services and EduHub.Data projects. Interfaces are placed to preserve dependency direction and facilitate DI; exact method signatures should be validated with implementation teams.

COVERAGE

Persistent class index

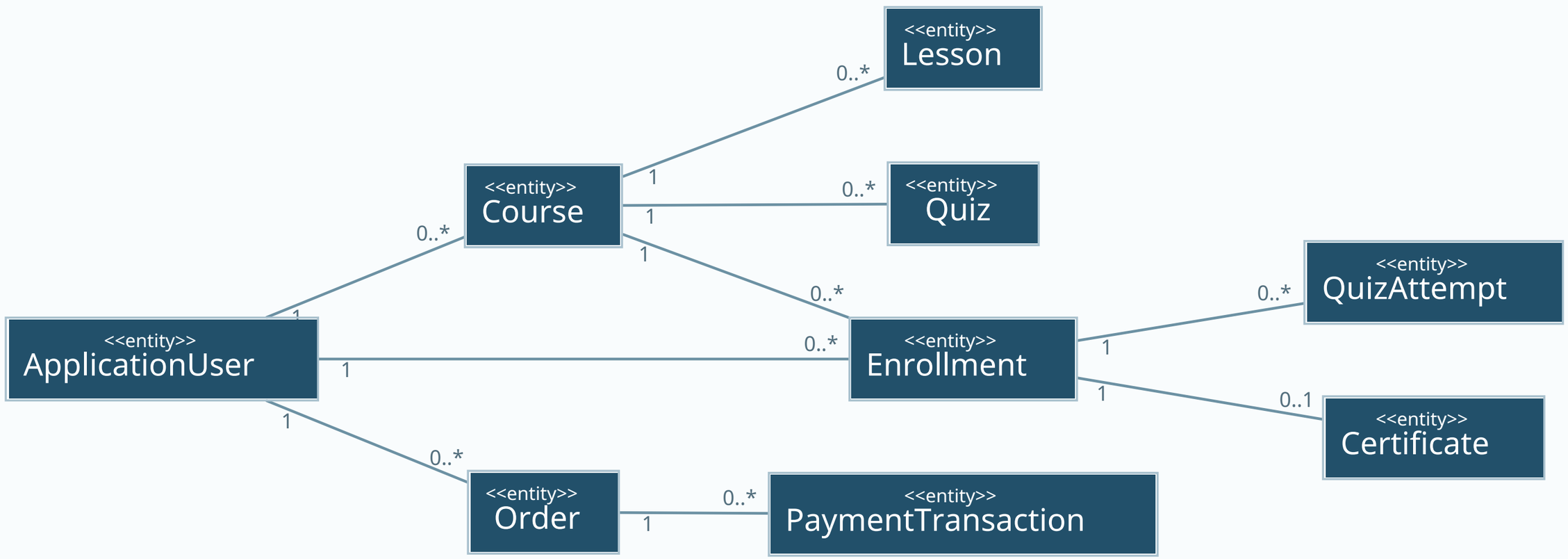
The following class names are the exact conceptual entities from the ERD, grouped by domain and cross-referenced to class views.

DOMAIN	CLASS NAMES	UML VIEWS
Identity (5)	ApplicationUser, ApplicationRole, UserRole, StudentProfile, InstructorProfile	CD-02
Catalog (7)	Category, Course, CourseSection, Lesson, MediaAsset, Tag, CourseTag	CD-03
Governance (2)	CourseApproval, CourseAnnouncement	CD-04
Learning (4)	Enrollment, LessonProgress, LessonNote, LessonBookmark	CD-05
Assessment (5)	Quiz, QuizQuestion, QuizOption, QuizAttempt, QuizAnswer	CD-06 / CD-07
Certification (1)	Certificate	CD-08
Commerce (9)	Cart, CartItem, Order, OrderItem, PaymentTransaction, PaymentWebhookEvent, Coupon, CouponRedemption, InstructorEarning	CD-09 / CD-10 / CD-11
Engagement (5)	CourseReview, ReviewHelpfulVote, CourseQuestion, CourseAnswer, Notification	CD-12

CD-01 / OVERVIEW

Domain model at a glance

Cross-domain structural relationships



MODEL STATUS / Persistent attributes trace to prior schema; visible operations and service classes are candidate design.

CD-01 / OVERVIEW

Domain model at a glance

Purpose, relationships and implementation constraints for the preceding vector diagram.

Purpose. A navigational class-level map of the platform. It intentionally omits junction and supporting classes; later diagrams restore the exact ERD pathways.

KEY CLASS RELATIONSHIPS

01 Courses are owned by instructors; students connect to courses through Enrollment, not a direct many-to-many property.

02 Enrollment is the anchor for learning records and certificates.

03 Orders and payment transactions are separate concepts; payment confirmation controls paid-course fulfillment.

IMPLEMENTATION & VALIDATION NOTES

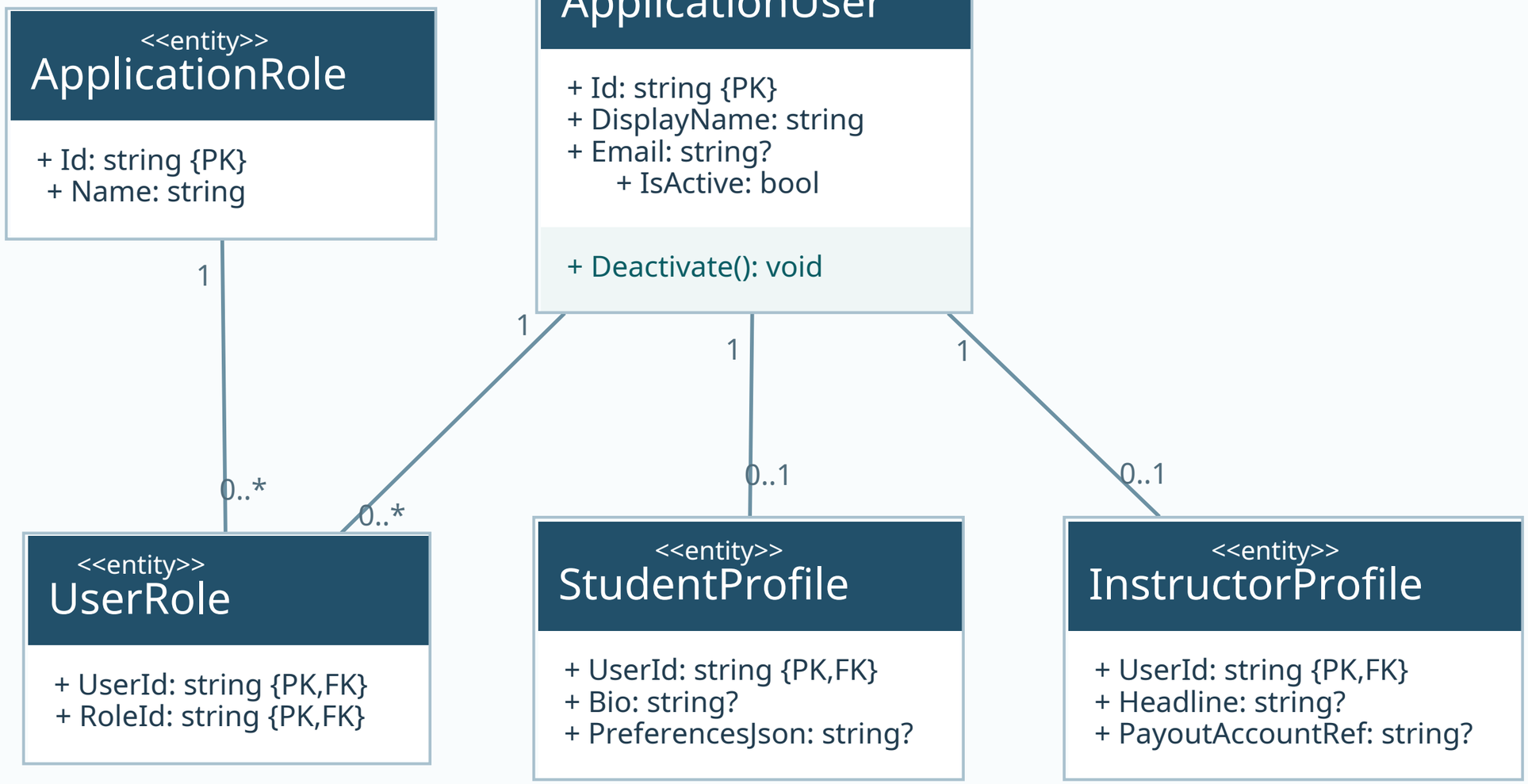
- Overview links can abbreviate indirect paths; CD-02 through CD-12 provide persistence-accurate relationships.
- Associations express domain relationships, not SQL cascades.

TRACEABILITY S1 S2 S3 | Sources identified in the bibliography

CD-02 / IDENTITY

Identity and roles

ASP.NET Core Identity with profile extensions



MODEL STATUS / Persistent attributes trace to prior schema; visible operations and service classes are candidate design.

CD-02 / IDENTITY

Identity and roles

Purpose, relationships and implementation constraints for the preceding vector diagram.

Purpose. Shared account identity with multiple role memberships; profile extensions are one-to-zero-or-one.

KEY CLASS RELATIONSHIPS

01 ApplicationUser represents the extended AspNetUsers principal.

02 UserRole is the explicit Identity join; users may hold multiple roles.

03 StudentProfile and InstructorProfile use UserId as both PK and FK.

IMPLEMENTATION & VALIDATION NOTES

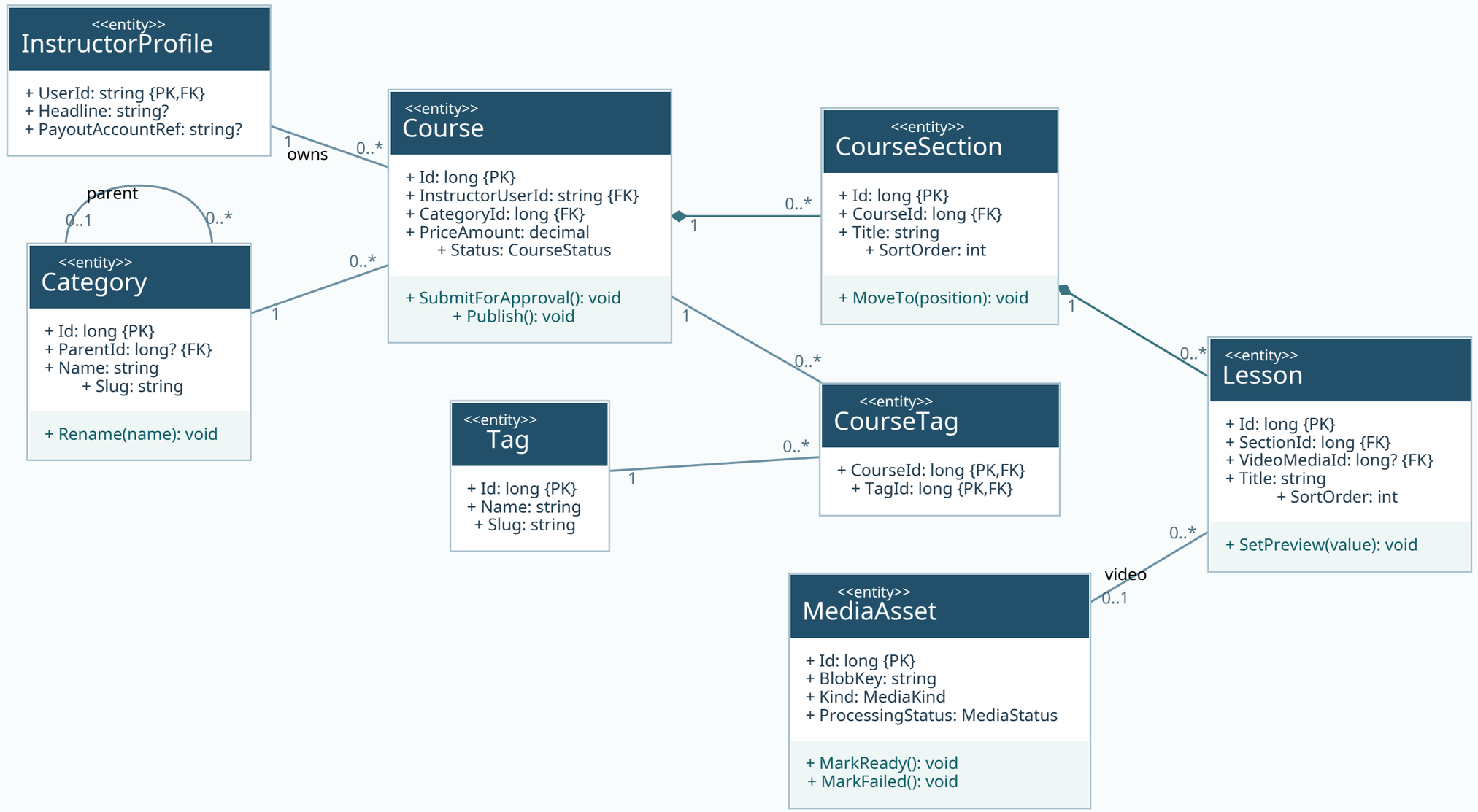
- Identity framework stores claims, logins and tokens outside the 38-entity conceptual count.
- An administrator is a role, not a third profile entity.

TRACEABILITY S1 S2 S5 | Sources identified in the bibliography

CD-03 / CATALOG

Course catalog and structure

Author-owned course hierarchy and taxonomies



MODEL STATUS / Persistent attributes trace to prior schema; visible operations and service classes are candidate design.

CD-03 / CATALOG

Course catalog and structure

Purpose, relationships and implementation constraints for the preceding vector diagram.

Purpose. Course content has ordered sections and lessons, with reusable media references and tag junctions.

KEY CLASS RELATIONSHIPS

01 A Course must have an InstructorProfile and Category.

02 Lessons require one section and may reference one video asset.

03 Category is hierarchical via nullable ParentId; CourseTag resolves many-to-many tags.

IMPLEMENTATION & VALIDATION NOTES

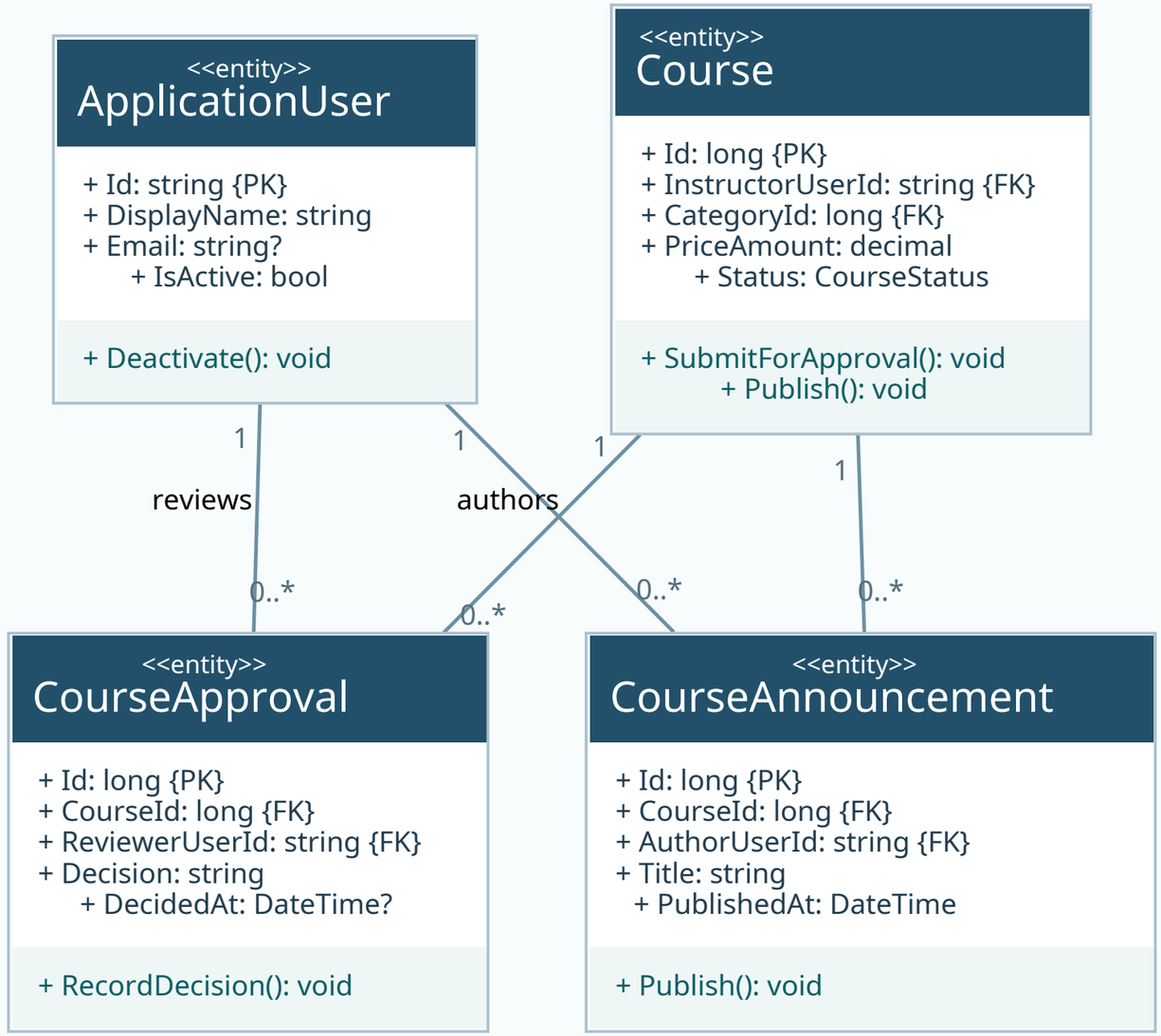
- Course.ThumbnailMediaId optionally references MediaAsset but is omitted for density; it remains part of the schema.
- Composition indicates conceptual ownership only; the SQL design uses controlled NO ACTION deletes.

TRACEABILITY S1 S2 | Sources identified in the bibliography

CD-04 / GOVERNANCE

Content governance

Administrative review and instructor communication



MODEL STATUS / Persistent attributes trace to prior schema; visible operations and service classes are candidate design.

CD-04 / GOVERNANCE

Content governance

Purpose, relationships and implementation constraints for the preceding vector diagram.

Purpose. Administrative decisions are retained as review records; announcements remain course-scoped.

KEY CLASS RELATIONSHIPS

01 CourseApproval stores reviewer, decision and decision date.

02 Course status and approval history are separate: the latest decision does not replace past audit records.

03 CourseAnnouncement records its author; publishing a notice does not change course approval.

IMPLEMENTATION & VALIDATION NOTES

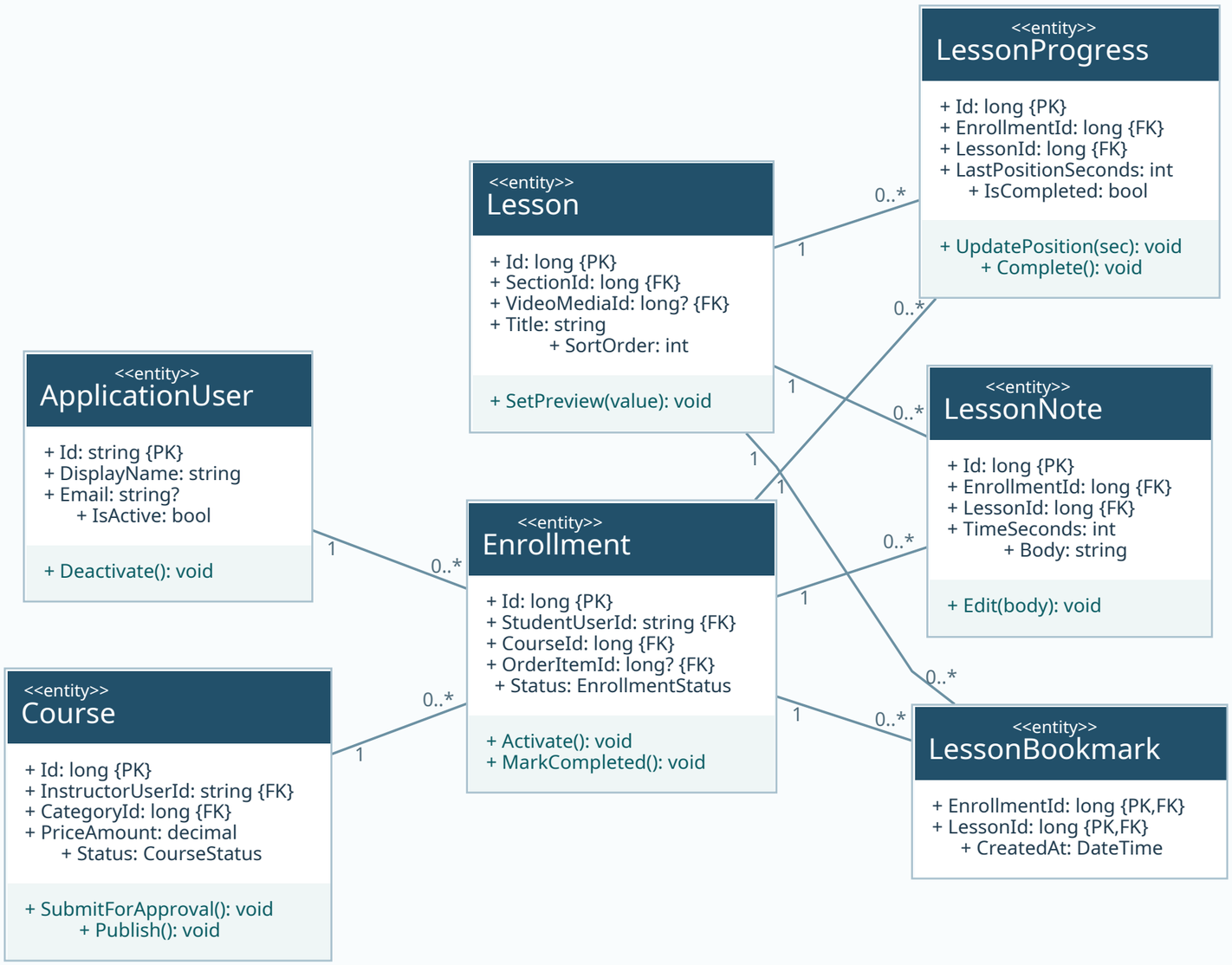
- Require Admin authorization for approval actions.
- Publication transition must follow the state machine rules from SM-01.

TRACEABILITY S1 S2 S4 | Sources identified in the bibliography

CD-05 / LEARNING

Enrollment and learning

Progress, student notes, and bookmarks



MODEL STATUS / Persistent attributes trace to prior schema; visible operations and service classes are candidate design.

CD-05 / LEARNING

Enrollment and learning

Purpose, relationships and implementation constraints for the preceding vector diagram.

Purpose. Enrollment defines the learner-course boundary; progress and notes reference both enrollment and lesson.

KEY CLASS RELATIONSHIPS

01 A unique pair (StudentUserId, CourseId) prevents duplicate enrollment.

02 A unique (EnrollmentId, LessonId) identifies each lesson progress row.

03 Bookmarks use a composite key; notes allow multiple timestamped entries for one lesson.

IMPLEMENTATION & VALIDATION NOTES

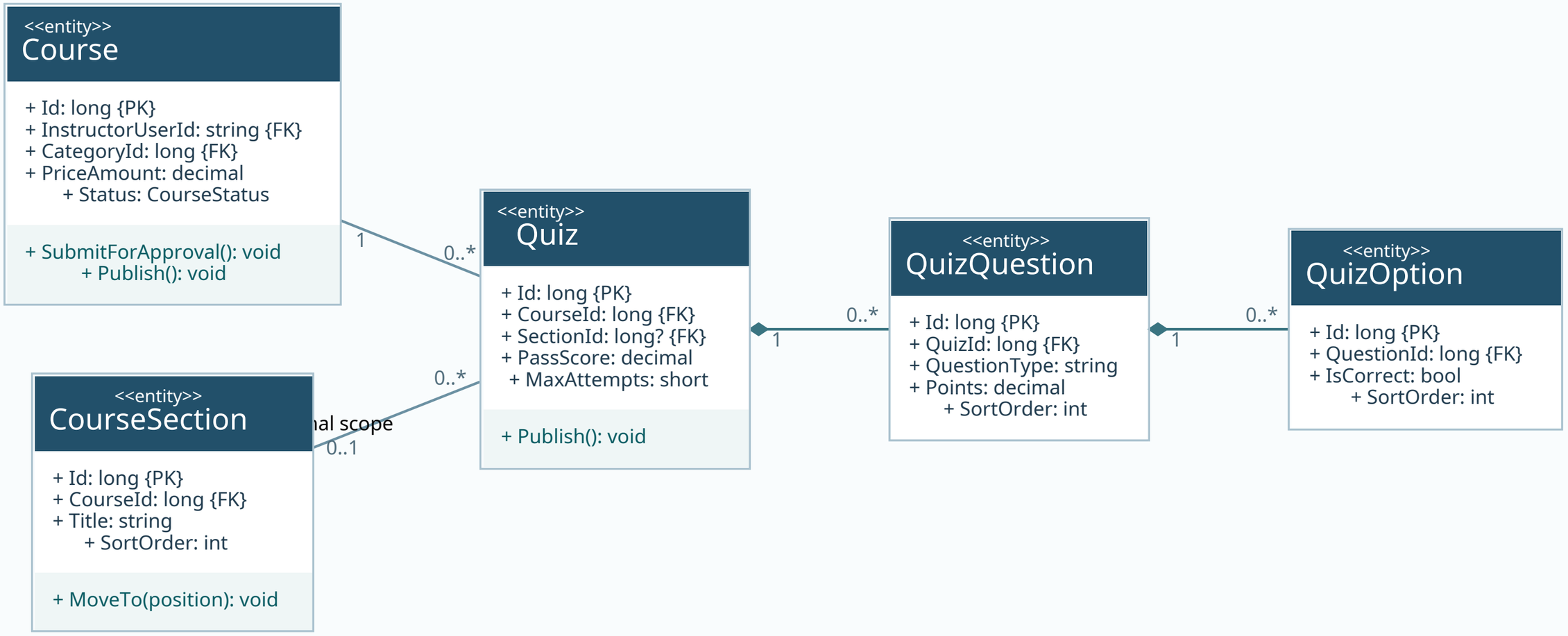
- Cross-course consistency (lesson belongs to enrollment course) needs transactional service validation.
- Enrollment.OrderItemId is optional for free courses.

TRACEABILITY S1 S2 S3 | Sources identified in the bibliography

CD-06 / ASSESSMENT

Quiz authoring structure

Definition classes for course and section assessments



MODEL STATUS / Persistent attributes trace to prior schema; visible operations and service classes are candidate design.

CD-06 / ASSESSMENT

Quiz authoring structure

Purpose, relationships and implementation constraints for the preceding vector diagram.

Purpose. Quiz definitions attach to a course and optionally a section; questions own their answer options.

KEY CLASS RELATIONSHIPS

01 Quiz.SectionId is nullable for course-level assessments.

02 QuizQuestion.QuestionType distinguishes MCQ, true/false and fill-in-the-blank.

03 CorrectText supports textual answers; QuizOption supports choice questions.

IMPLEMENTATION & VALIDATION NOTES

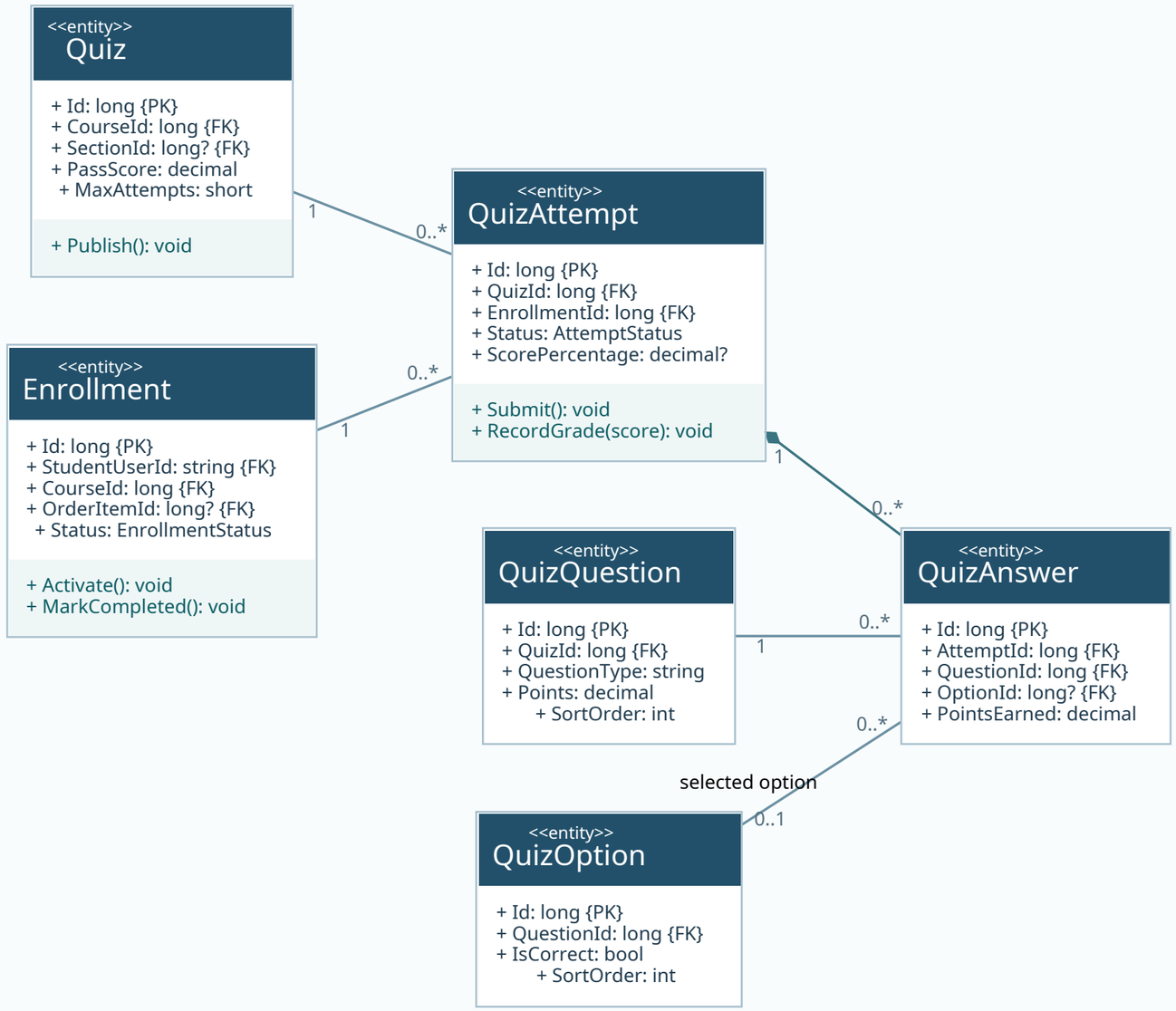
- The quiz course must equal the parent course of any referenced section.
- MaxAttempts and PassScore belong to Quiz, not to the attempt record.

TRACEABILITY S1 S2 | Sources identified in the bibliography

CD-07 / ASSESSMENT

Quiz attempts and answers

Learner submissions, grading and evidence



MODEL STATUS / Persistent attributes trace to prior schema; visible operations and service classes are candidate design.

CD-07 / ASSESSMENT

Quiz attempts and answers

Purpose, relationships and implementation constraints for the preceding vector diagram.

Purpose. Attempts preserve attempt number, score and submitted answers instead of overwriting a single quiz result.

KEY CLASS RELATIONSHIPS

01 QuizAttempt belongs to one quiz and one enrollment.

02 QuizAnswer belongs to a question from that attempt quiz.

03 OptionId is optional for text-answer question types.

IMPLEMENTATION & VALIDATION NOTES

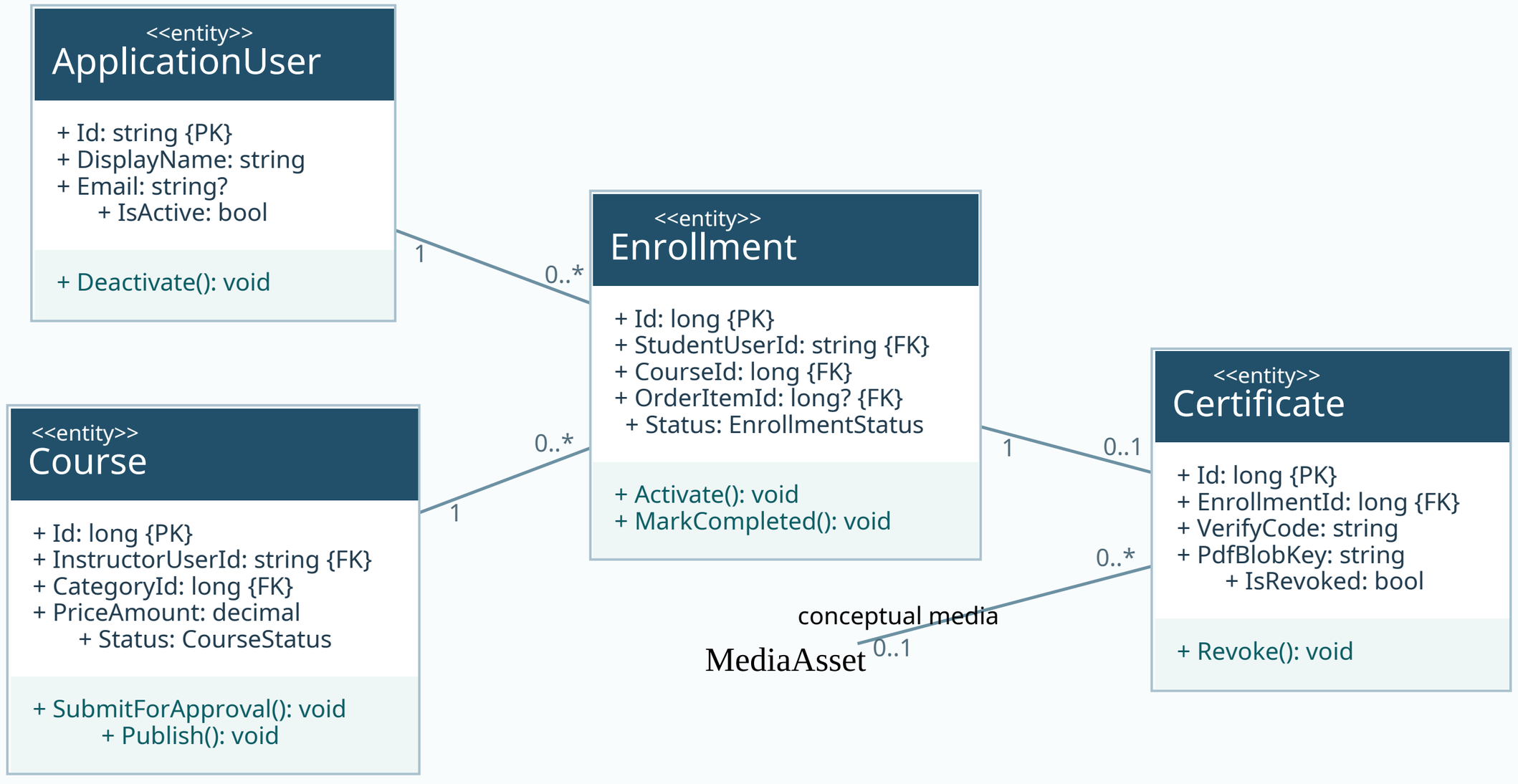
- The course of Quiz and the course of Enrollment must match.
- At most one answer per question within an attempt; the previous schema enforces the uniqueness constraint.

TRACEABILITY S1 S2 S3 | Sources identified in the bibliography

CD-08 / CERTIFICATION

Certificates and verification

Issued credentials linked to course completion



MODEL STATUS / Persistent attributes trace to prior schema; visible operations and service classes are candidate design.

CD-08 / CERTIFICATION

Certificates and verification

Purpose, relationships and implementation constraints for the preceding vector diagram.

Purpose. Certificate records are attached one-to-one (optional) to an eligible enrollment and expose a unique public verification code.

KEY CLASS RELATIONSHIPS

01 Issuance requires completion and passing policy checks.

02 VerificationCode is unique; revocation is modeled separately from PDF generation.

03 PDF Blob location is stored as Certificate.PdfBlobKey in the schema.

IMPLEMENTATION & VALIDATION NOTES

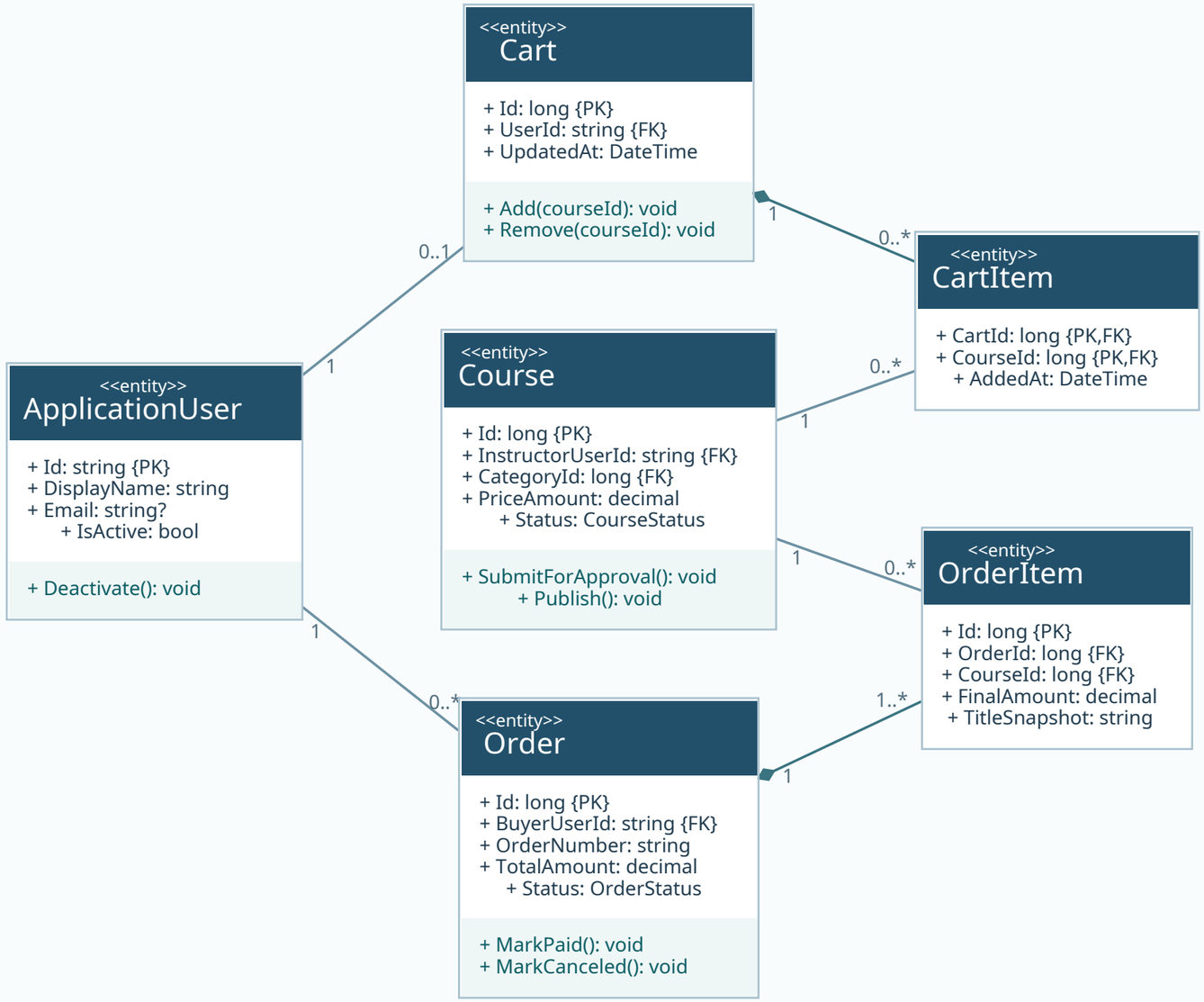
- Certificate.PdfBlobKey is a scalar Azure Blob reference; no MediaAsset association or foreign key exists in the physical schema.
- Do not infer certificate validity from a PDF file alone.

TRACEABILITY S1 S2 S4 | Sources identified in the bibliography

CD-09 / COMMERCE

Cart and order structure

Shopping basket and immutable purchase snapshots



MODEL STATUS / Persistent attributes trace to prior schema; visible operations and service classes are candidate design.

CD-09 / COMMERCE

Cart and order structure

Purpose, relationships and implementation constraints for the preceding vector diagram.

Purpose. Cart lines are mutable selections; OrderItem is a financial snapshot at checkout.

KEY CLASS RELATIONSHIPS

01 CartItem is the composite-key junction (CartId, CourseId).

02 OrderItem preserves title and final price snapshot even if Course changes.

03 An Order contains one or more items in a valid placed checkout.

IMPLEMENTATION & VALIDATION NOTES

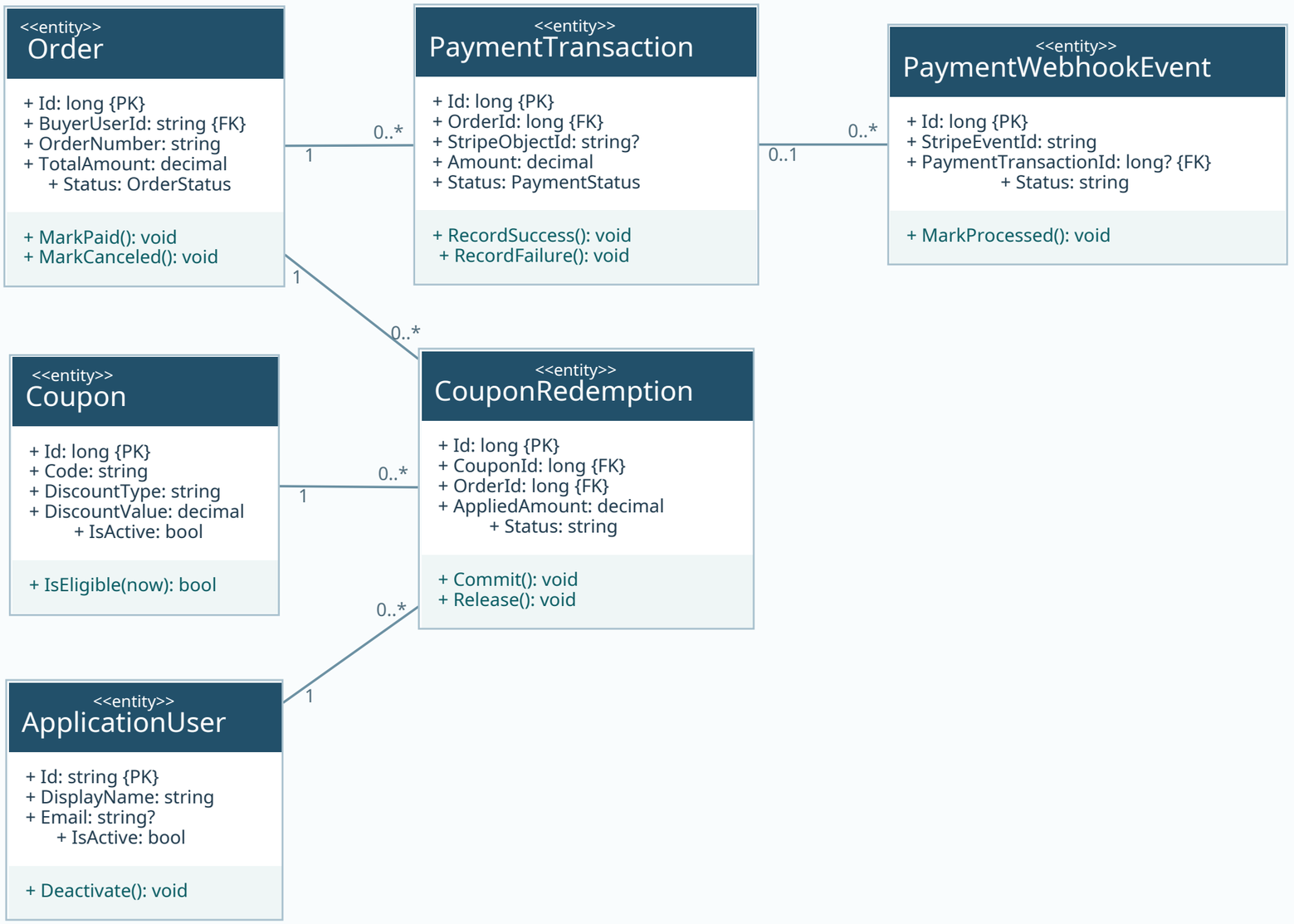
- A database cannot express the one-or-more order item rule with a simple FK; enforce in service transaction.
- Cart items do not themselves grant enrollment.

TRACEABILITY S1 S2 S3 | Sources identified in the bibliography

CD-10 / COMMERCE

Payments and promotions

Gateway transactions, webhook deduplication and coupon usage



MODEL STATUS / Persistent attributes trace to prior schema; visible operations and service classes are candidate design.

CD-10 / COMMERCE

Payments and promotions

Purpose, relationships and implementation constraints for the preceding vector diagram.

Purpose. Payment and promotion records are separated to support retries, immutable history and Stripe webhook idempotency.

KEY CLASS RELATIONSHIPS

01 An Order may have several payment attempts; each PaymentTransaction belongs to one Order.

02 Webhook events have globally unique provider event IDs.

03 CouponRedemption records the applied monetary discount for an order.

IMPLEMENTATION & VALIDATION NOTES

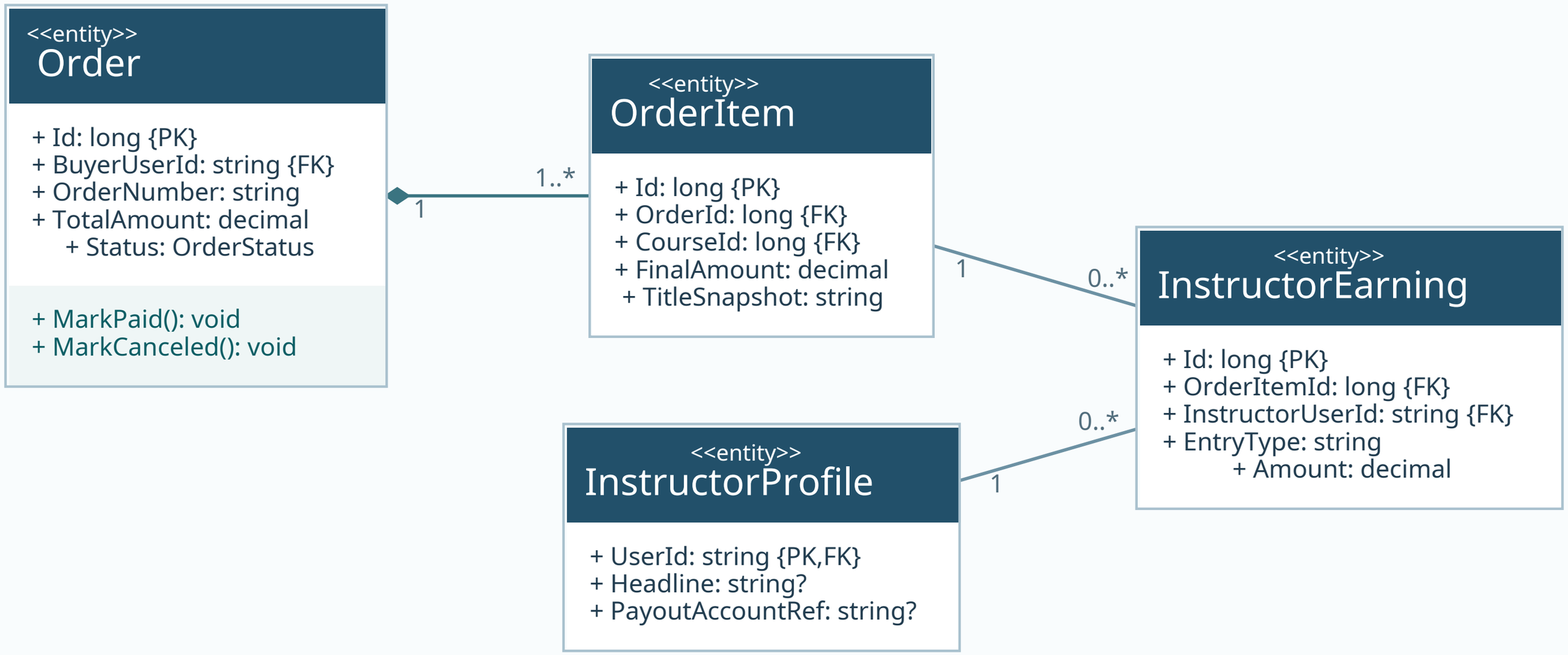
- PaymentWebhookEvent.PaymentTransactionId may be null until an event is correlated.
- Successful enrollment fulfillment requires a verified server-side Stripe signal, not only a browser redirect.

TRACEABILITY S1 S2 S3 | Sources identified in the bibliography

CD-11 / COMMERCE

Instructor earnings ledger

Revenue recognition based on purchased course items



MODEL STATUS / Persistent attributes trace to prior schema; visible operations and service classes are candidate design.

CD-11 / COMMERCE

Instructor earnings ledger

Purpose, relationships and implementation constraints for the preceding vector diagram.

Purpose. Earnings are ledger-like records tied to purchased course lines, supporting positive entries and reversals.

KEY CLASS RELATIONSHIPS

01 InstructorEarning.OrderItemId is mandatory.

02 InstructorEarning.InstructorUserId identifies the earning recipient.

03 EntryType and EventRef distinguish payout-related adjustments or reversals.

IMPLEMENTATION & VALIDATION NOTES

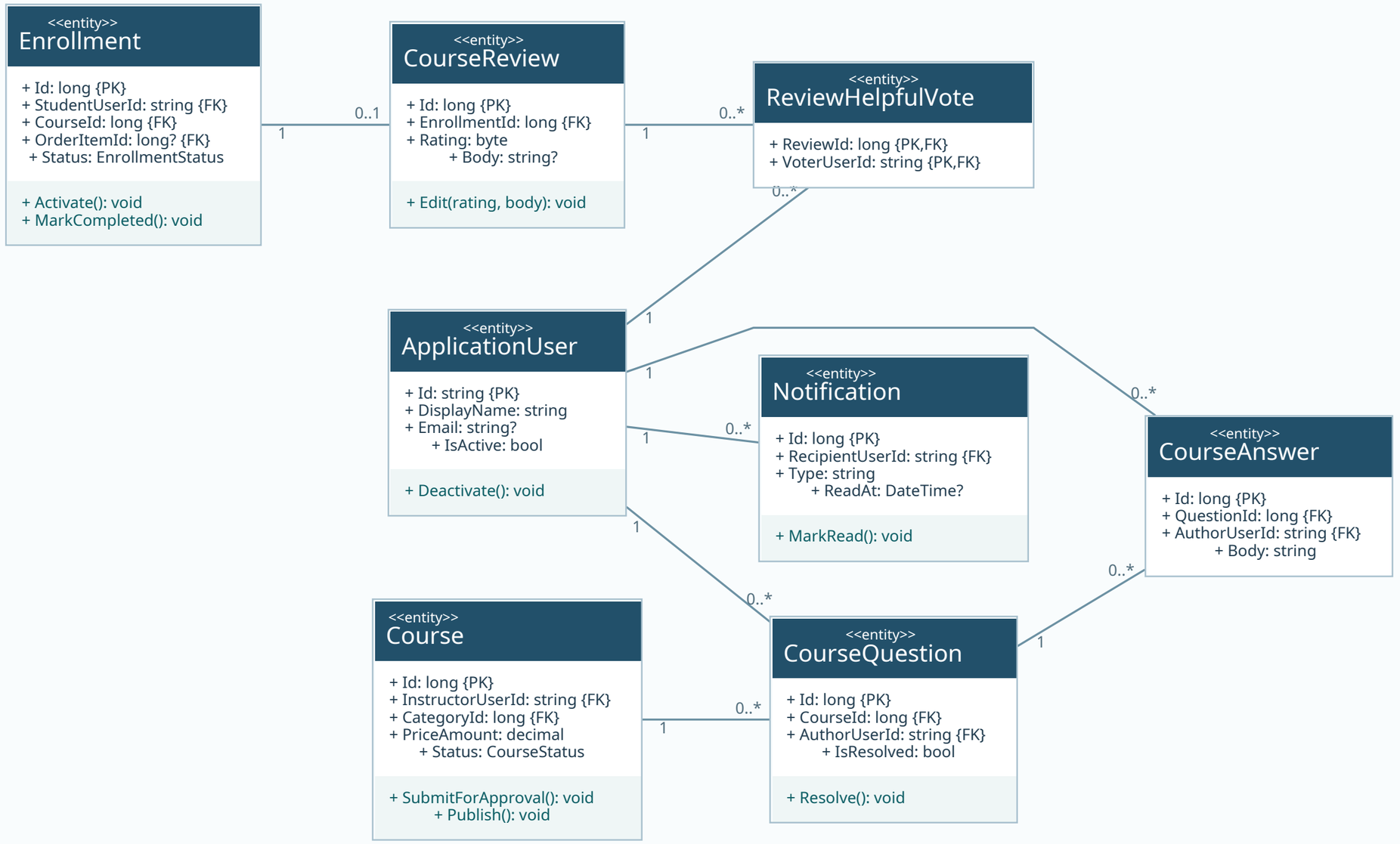
- Earning generation belongs after paid-order fulfillment and requires duplicate safeguards.
- Direct instructor payout orchestration is not specified in the brief.

TRACEABILITY S1 S2 S3 | Sources identified in the bibliography

CD-12 / ENGAGEMENT

Engagement and messaging

Reviews, helpful votes, questions, answers and notifications



MODEL STATUS / Persistent attributes trace to prior schema; visible operations and service classes are candidate design.

CD-12 / ENGAGEMENT

Engagement and messaging

Purpose, relationships and implementation constraints for the preceding vector diagram.

Purpose. Each enrollment can contribute one review; students ask questions and instructors answer; notifications reference their recipient.

KEY CLASS RELATIONSHIPS

01 CourseReview.EnrollmentId is unique for one review per enrollment.

02 Helpful votes have composite ReviewId/VoterUserId keys.

03 CourseQuestion and CourseAnswer retain author IDs for audit.

IMPLEMENTATION & VALIDATION NOTES

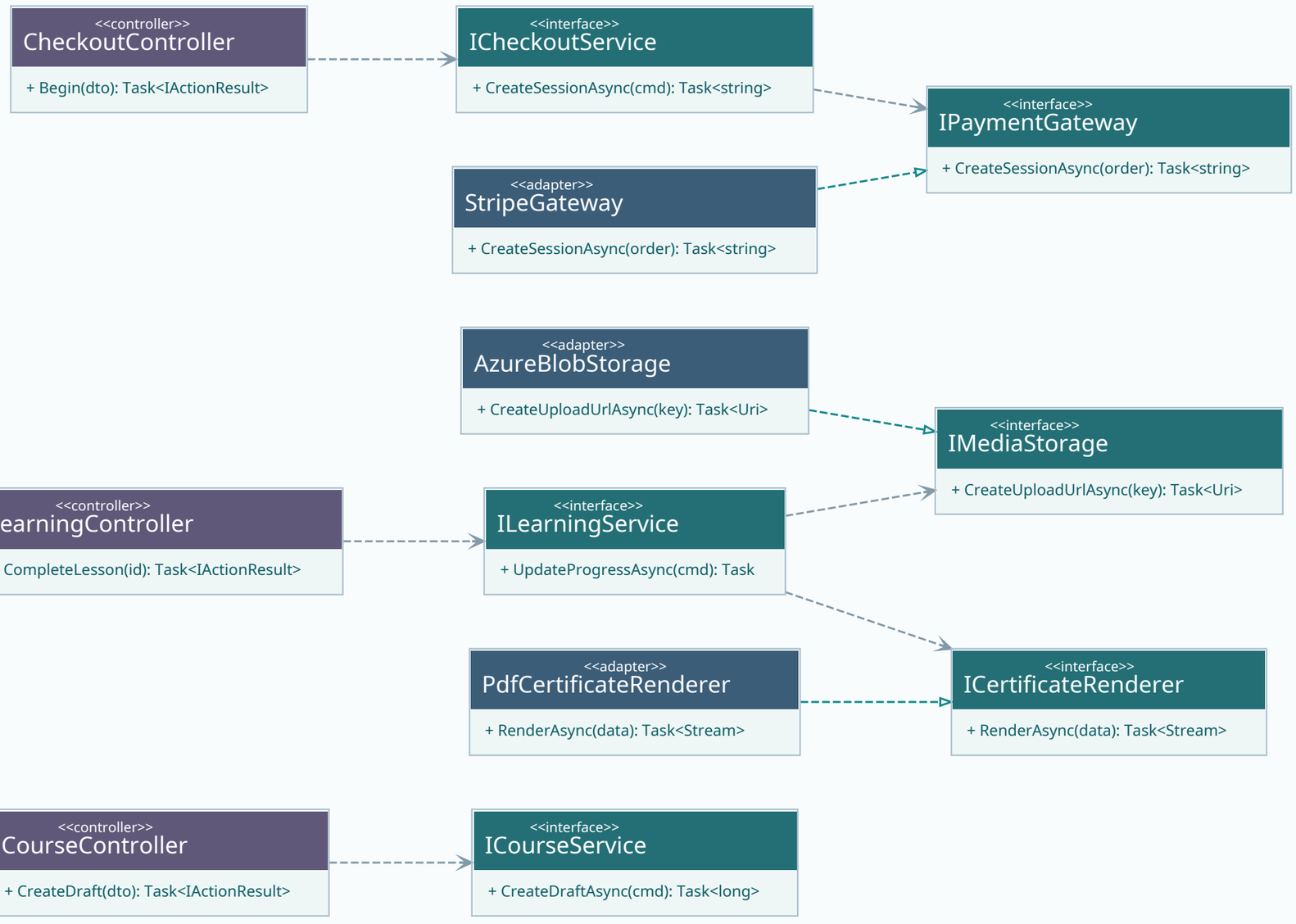
- Role/ownership requirements for answering questions are enforced in services, not FKs.
- Notifications can be triggered by progress events using the Observer pattern.

TRACEABILITY S1 S2 S3 | Sources identified in the bibliography

CD-13 / APPLICATION

Application-layer contracts

MVC controller to service and infrastructure interfaces



MODEL STATUS / Persistent attributes trace to prior schema; visible operations and service classes are candidate design.

CD-13 / APPLICATION

Application-layer contracts

Purpose, relationships and implementation constraints for the preceding vector diagram.

Purpose. Technical design-level classes show direction of dependency across the three required assemblies.

KEY CLASS RELATIONSHIPS

01 EduHub.Web contains controllers; EduHub.Services owns use-case interfaces and orchestration.

02 EduHub.Data implements EF Core persistence and third-party adapters.

03 Controllers receive abstractions via dependency injection and never issue SQL commands.

IMPLEMENTATION & VALIDATION NOTES

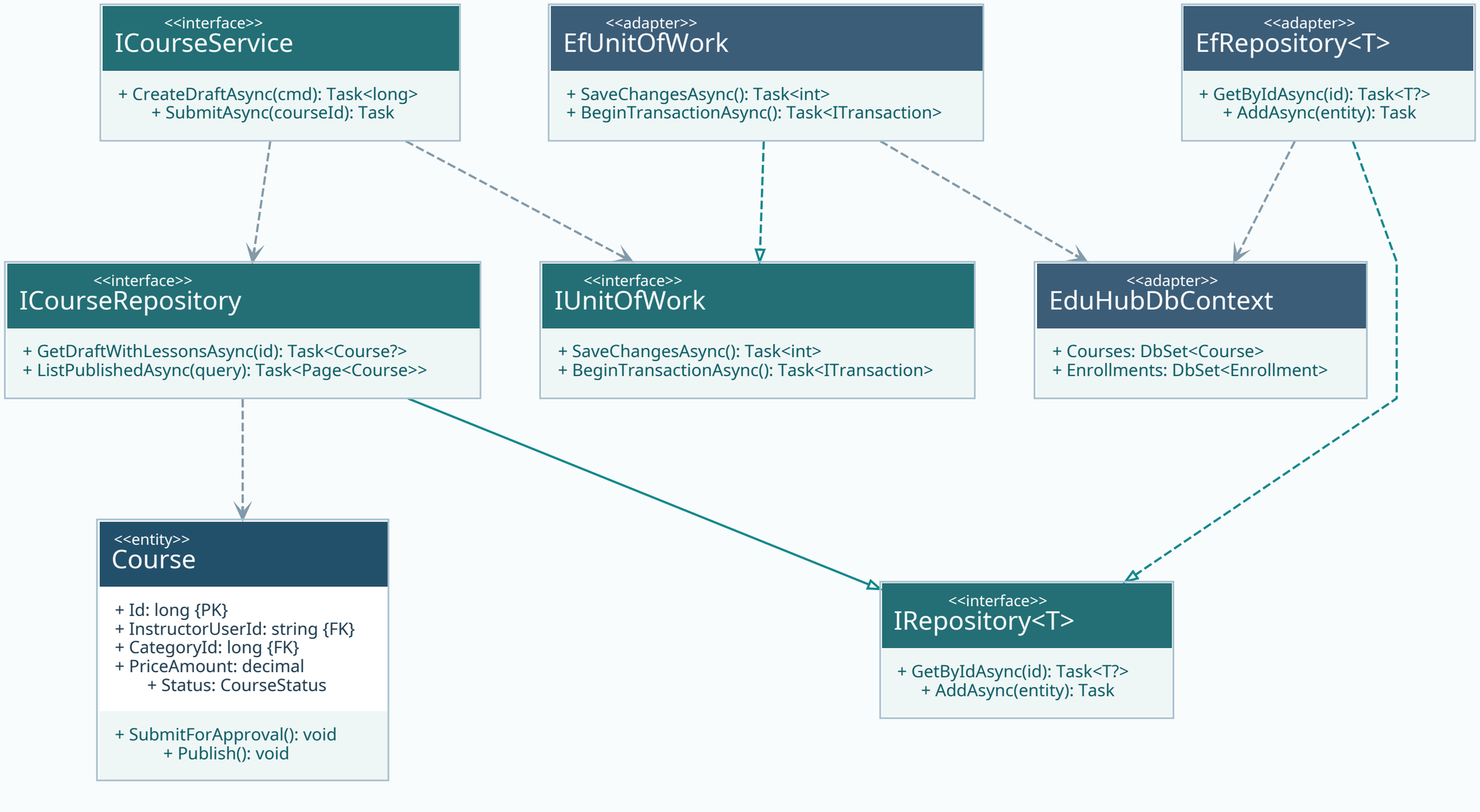
- Service and adapter class names are proposed; the source brief fixes layer boundaries but not these specific interface names.
- Do not equate interface arrows with persisted relationships.

TRACEABILITY S1 S3 S5 | Sources identified in the bibliography

CD-14 / PERSISTENCE

Repository and Unit of Work

Persistence interfaces and EF Core implementations



MODEL STATUS / Persistent attributes trace to prior schema; visible operations and service classes are candidate design.

CD-14 / PERSISTENCE

Repository and Unit of Work

Purpose, relationships and implementation constraints for the preceding vector diagram.

Purpose. Repository interfaces abstract queries; Unit of Work coordinates committing related entity changes.

KEY CLASS RELATIONSHIPS

01 IRepository<T> provides shared CRUD-like operations.

02 ICourseRepository adds domain-specific read queries.

03 The DbContext is the EF Core unit of work underneath the proposed abstraction.

IMPLEMENTATION & VALIDATION NOTES

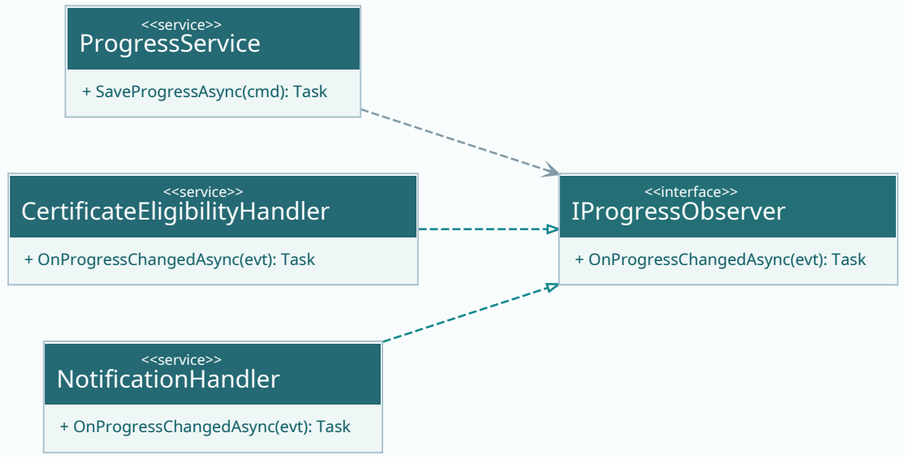
- ICourseRepository specializes the IRepository<T> template with T = Course; one SaveChanges call should close the transaction.
- The source brief explicitly requires Repository and Unit of Work; exact contract shapes are proposals.

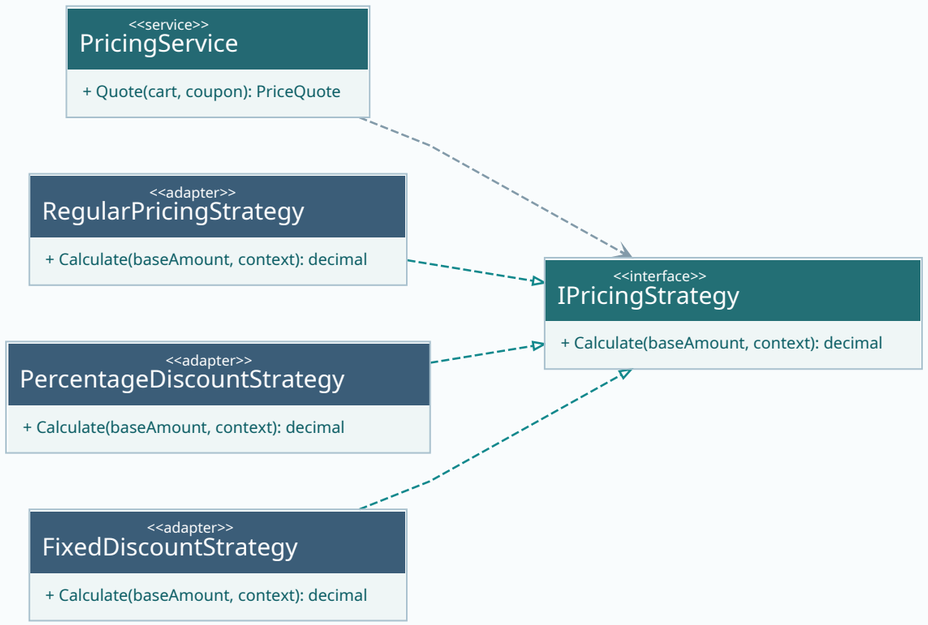
TRACEABILITY S1 S3 S5 | Sources identified in the bibliography

CD-15 / PATTERNS

Strategy and Observer patterns

Extensible pricing rules and event-driven learning reactions





MODEL STATUS / Persistent attributes trace to prior schema; visible operations and service classes are candidate design.

CD-15 / PATTERNS

Strategy and Observer patterns

Purpose, relationships and implementation constraints for the preceding vector diagram.

Purpose. Strategy chooses discount calculation behavior; Observer reacts to committed progress events.

KEY CLASS RELATIONSHIPS

01 IPricingStrategy makes percentage/fixed discount calculations replaceable.

02 ProgressService publishes progress-changed events only after data is committed.

03 Eligibility and notification handlers are subscribers; duplicate effects must be safe.

IMPLEMENTATION & VALIDATION NOTES

- Observer is a conceptual design: implementation could use in-process domain events and an outbox if reliability needs increase.
- No separate message broker or microservice is required.

TRACEABILITY S1 S3 S5 | Sources identified in the bibliography

ENGINEERING NOTES

Class boundaries are not table boundaries

Avoid treating the database schema as a one-to-one blueprint for every technical object.

AREA	PROPOSED IMPLEMENTATION RULE	REASON / TEST
Entity state	Use behavior methods as centralized guards, or equivalent service-layer checks; never update sensitive statuses directly from MVC controllers.	Exercise forbidden course, enrollment, payment and certificate transitions.
Identity	Extend IdentityUser via ApplicationUser; do not create duplicate credential tables.	Confirm IdentityDbContext migrations own security tables.
Navigation	Explicit FK + nullable reference and collection navigations as needed, configured with Fluent API.	Cross-check multiplicity against SQL FK nullability.
Repository / UoW	One DbContext per scoped operation; commit once per transactional business action.	Test rollback for enrollment + payment fulfillment.
Payments	Separate checkout request, Stripe transaction, webhook receipt and entitlement update.	Deduplicate StripeEventId; ensure replay has no duplicate enrollment.
Progress	Persist change first, then invoke observers via reliable post-commit dispatch.	Repeated progress event does not duplicate credential.
Media	Reference blob keys; signed URLs are generated by storage adapter, not persisted credentials.	Check expired/read-only token behavior.

DESIGN TRACEABILITY

Class relationships to implementation evidence

This is a review matrix, not a claim that the application has been coded or tested.

DESIGN SUBJECT	CLASS MODEL	BASELINE / ACCEPTANCE CHECK
Role membership and profiles	CD-02	ERD Identity (5 entities), Identity model, FR-001 to FR-006.
Catalog structure, governance	CD-03 / CD-04	Course / Section / Lesson FKs, CourseStatus and admin approval.
Enrollment and lesson tracking	CD-05	Unique student-course + lesson progress; FR-030 onward.
Quiz lifecycle	CD-06 / CD-07	Section vs course quiz, attempt number, option validation.
Credential issuance	CD-08	Enrollment 0..1 Certificate; verify code unique; revocation rules.
Checkout and payments	CD-09 / CD-10 / CD-11	Order snapshot, Stripe event dedup, paid-order fulfillment, earning ledger.
Reviews and communication	CD-12	Review uniqueness, helpful votes, Q&A; authorship and notifications.
Layer and patterns	CD-13 / CD-14 / CD-15	N-Tier, Repository, Unit of Work, Strategy, Observer and Dependency Injection.

VALIDATION

Architecture and model acceptance checklist

Resolve these questions during the System Analysis & Design review before development.

☐ **Entity inventory**	All 38 entity classes are present in domain diagrams CD-02 through CD-12.
☐ **Keys and optionals**	Every optional FK is indicated as nullable; composite junction PKs are preserved.
☐ **Navigations**	Each EF Core relationship has compatible FK and navigation definitions.
☐ **User roles**	Course publishing and approval checks are authorized at service boundaries.
☐ **Cross-entity guards**	Quiz/course/enrollment and lesson/course/enrollment compatibility is verified.
☐ **Payments**	Webhook processing is signature-validated, idempotent, and transaction-safe.
☐ **Media**	Private blob URLs are issued only after entitlement authorization.
☐ **Certificates**	Eligibility and one-per-enrollment rule are enforced transactionally.
☐ **Lifecycle**	Domain state mutation is consistent with previous State Diagrams.
☐ **Patterns**	DI lifetime of DbContext/UoW is scoped; no singleton captures scoped service.
☐ **Scope signoff**	Proposed member methods and interface names are accepted or documented as alternatives.

SOURCE CONTROL

Document sources and references

The supplied project brief and previously authored EduHub documents are the design baseline; external sources define UML and framework conventions.

ID	SOURCE	USAGE / SCOPE	LINK OR LOCATION
S1	EduHub_ProjectDetails(1).pdf	Project 3: EduHub - Online Learning Platform, 5 pages; objective, modules, N-Tier projects and required patterns.	Local project attachment
S2	EduHub_Logical_Physical_Schema.pdf; EduHub_SQLServer_Schema.sql	Prior 38-entity relational model and full proposed column/constraint definitions.	Prior EduHub deliverables
S3	EduHub_Software_Architecture.pdf; EduHub_Sequence_Diagrams.pdf	Layer boundaries, payment fulfillment, video access and application interaction flows.	Prior EduHub deliverables
S4	EduHub_State_Diagrams.pdf	Entity lifecycle, persistence states and proposed transition guards.	Prior EduHub deliverable
S5	OMG Unified Modeling Language 2.5.1	UML class representation and association / realization / generalization conventions.	https://www.omg.org/spec/UML/2.5.1/
S6	Microsoft Learn - EF Core Relationships	Relationship navigations and FK mapping conventions.	https://learn.microsoft.com/en-us/ef/core/modeling/relationships
S7	Microsoft Learn - ASP.NET Core Dependency Injection	Dependency injection container and service lifetime considerations.	https://learn.microsoft.com/en-us/aspnet/core/fundamentals/dependency-injection

Design fidelity: No external reference is treated as a source of project-specific features. New service interfaces, domain methods and technical adapters are proposed engineering design choices, not statements that code currently exists.